

Activity 1: Understanding the Project with Generative AI

Purpose

Before developing any software application, developers must understand the business problem, application scope, users, and functional requirements. In this activity, you will use Generative AI to analyze the SmartBank Net Banking Application and gain a high-level understanding of the project.

Pre-requisites

- ChatGPT or Oracle Chat AI access
 - Internet connection
 - Basic knowledge of software applications
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Estimated Duration

30 Minutes

Scenario

You have recently joined **ABC Technologies** as a Graduate Software Engineer.

Your first assignment is to work on **SmartBank**, a Net Banking Application. Before writing any code, your Team Lead has asked you to understand the project, identify its major components, and become familiar with the application.

Use Generative AI as your learning assistant to explore the project.

Tasks

Complete the following tasks in sequence.

Task 1: Understand the Business Problem

Use AI to understand the purpose of the SmartBank application, its users, business objectives, and major features.

Prompt

Act as a Senior Software Architect.

I have recently joined a software development company as a Graduate Software Engineer.

Our team is developing a Net Banking Application called SmartBank.

The application should allow customers to:

- Login securely
- View account balances
- Transfer funds
- Manage beneficiaries
- Pay utility bills
- View transaction history
- Download account statements
- Update their profile

Explain the project in simple English.

Include:

1. Purpose of the application
2. Business problem it solves
3. Primary users
4. Major features
5. Benefits for customers

Present your response using headings and bullet points.

Task 2: Identify the Major Functional Modules

Break the application into logical business modules.

Prompt

Based on the SmartBank project, identify the major functional modules.

For each module provide:

- Module Name
- Purpose
- Main Features
- Business Importance

Present the information as a table.

Task 3: Identify the Users

Identify the different users who interact with the application.

Prompt

Identify all users of the SmartBank application.

For each user provide:

- User Role
- Responsibilities
- Features they will access

Present the information as a table.

Task 4: Explore the Technology Stack

Use AI to understand the technologies commonly used for enterprise banking applications.

Prompt

Suggest a suitable technology stack for developing the SmartBank application.

Include:

- Frontend
- Backend
- Database
- Authentication
- REST API
- Testing
- Version Control
- CI/CD

Briefly explain the purpose of each technology.

Task 5: Understand the Project Team

Identify the different roles involved in software development.

Prompt

Explain the different roles involved in developing the SmartBank application.

Include:

- Business Analyst
- UI/UX Designer
- Frontend Developer
- Backend Developer
- Database Developer
- QA Engineer
- DevOps Engineer
- Project Manager

For each role explain:

- Primary responsibilities
 - Major deliverables
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Expected Deliverables

Upon completion of this activity, you should have:

- A high-level understanding of the SmartBank application.
 - A list of the application's functional modules.
 - A list of different users and their responsibilities.
 - An overview of the technology stack used in enterprise banking applications.
 - An understanding of the different roles involved in software development.
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Challenge Task

Imagine that another fresher has joined your project today and knows nothing about SmartBank.

Use AI to create a **one-page project overview document** that can be shared during project onboarding.

Prompt

Act as a Technical Lead.

Create a one-page project overview document for the SmartBank Net Banking Application.

The document should include:

- Project Overview
- Business Objectives
- Major Modules
- Primary Users
- Technology Stack
- Development Team Structure

The document should be simple, professional, and suitable for onboarding new software engineers.